Local Readout of Global Thermal Information

An additive information-loss theorem, a variance-only sensing certificate, and the limits of the global-conditioning route

ASTRA research derivation

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Abstract

For an unrestricted finite-temperature Gibbs family whose parameter directions are bounded-support, bounded-incidence Pauli interactions, we prove that one fixed randomized single-qubit measurement loses at most a size-independent matrix multiple of the coefficient Euclidean metric relative to symmetric-logarithmic-derivative quantum Fisher information. Equivalently, the loss in an extensive collective direction is at most extensive. The bound permits noncommuting interactions, frustration, arbitrary finite temperature, and calibrated independent detector sign errors. The background Hamiltonian only needs bounded support and incident weighted strength, not bounded combinatorial degree. The proof controls the *difference* between measured covariance and thermal response; it never assumes that thermal correlations decay.

The result implies that superextensive equilibrium parameter information, when present at fixed local parameters and temperature, is asymptotically accessible by fixed local readout and even by a scalar moment statistic. A measured scalar variance yields a two-sided certificate for optimal quantum information, with an explicit finite-copy confidence bound and no partition-function evaluation. These statements do not use the earlier ASTRA global covariance candidate. That candidate, if established, would additionally give an all-directions constant-factor comparison. Exact counterexamples separate the result from convex maximum-likelihood optimization, vanishing extensive readout loss, and robustness to correlated detector errors.

The elementary inequalities used here have established antecedents. The complete combination is derived below; independent proof review, original priority, and an end-to-end advantage over existing learning algorithms are not established.

1 Question, status, and the change of representation

The preceding ASTRA manuscript proposed a system-size-independent lower bound on centered KMS and Bogoliubov–Kubo–Mori (BKM) covariance for canonical Pauli interactions of bounded candidate incidence. Its exact-adjoint construction is an internally derived, unreviewed candidate, not a premise accepted here. Repository record N711 preserves that status and the independent-review gate.

The significant operational question is not merely whether local moments uniquely specify coefficients. It is whether a feasible, parameter-independent readout can access information which an ideal, possibly entangling measurement could extract. Lower BKM conditioning alone does not answer this question or supply an efficient likelihood evaluator.

The mechanism below changes the object to be bounded. Ordinary covariances may have long-range, phase-dependent contributions. But a local shadow estimator reproduces those contributions exactly between disjoint interaction supports. Its *extra* covariance is supported on the overlap graph. A thermal correlation inequality similarly bounds the noncommuting

difference between static variance and BKM response by a local double commutator. These two differences are bounded in coefficient operator norm without bounding the full covariance.

This gives an independent route to an operational theorem. The Gaussian thermal-adjoint argument is not needed for it.

2 Model, measurement, and information conventions

2.1 Parameter and background locality

Let $P_1, \dots, P_m$ be distinct canonical-sign, nonidentity Pauli strings on n qubits. Write $S_a = \operatorname{supp} P_a$ and suppose

$$|S_a| \le k, \qquad \max_i \#\{a : i \in S_a\} \le \Delta. \tag{1}$$

The bound is on the *parameter dictionary*, including zero-valued coefficients. Take

$$H(h) = H_{\mathrm{bg}} + \sum_{a=1}^{m} h_a P_a, \qquad \rho_h = \frac{e^{-\beta H(h)}}{\operatorname{Tr} e^{-\beta H(h)}}, \qquad \beta > 0. \tag{2}$$

The background is fixed with respect to h. At the operating point suppose the full Hamiltonian has a decomposition

$$H = \sum_X H_X, \qquad |X| \le \ell, \qquad \max_i \sum_{X \ni i} \|H_X\|_{\mathrm{op}} \le g. \tag{3}$$

No bound on the number of H_X meeting a site is necessary. Thus background interactions may be numerous and weak. There is no lattice, clustering, commutativity, gap, or mixing-time hypothesis. The state is the unrestricted full-rank Gibbs state, not a fixed-charge restriction or an arbitrary metastable state.

For real $u \in \mathbb{R}^m$, write $A_u = \sum_a u_a P_a$ and $q_u = \|u\|_2^2$. For an observable A, let $A_c = A - \operatorname{Tr}(\rho A) I$. Define real symmetric matrices C and V_q through

$$u^T C u = \int_0^1 \operatorname{Tr}(\rho^t (A_u)_c \rho^{1-t} (A_u)_c)\, dt, \tag{4}$$

$$(V_q)_{ac} = \tfrac{1}{2} \operatorname{Tr} \rho \{(P_a)_c, (P_c)_c\}. \tag{5}$$

The subscript q denotes ordinary symmetrized quantum covariance, not Fisher information.

2.2 A fixed local readout, including independent sign noise

On each independently prepared copy, choose at every site a basis $b_i \in \{X, Y, Z\}$ uniformly and independently, measure it, and record both the basis and the sign $s_i \in \{-1, +1\}$. Allow a parameter-independent sign-flip channel at site i with calibrated visibility $\eta_i \in [\eta, 1]$, where $\eta > 0$ is fixed. The sign error probability is $(1 - \eta_i)/2$; errors are independent across sites and copies. Quantum outcome correlations are *not* assumed independent.

The joint POVM, including the basis labels, is

$$E_{b,s} = \bigotimes_{i=1}^{n} \frac{I + \eta_i s_i \sigma_{b_i}}{6}. \tag{6}$$

For each parameter direction define the real feature

$$Y_a(b, s) = \prod_{i \in S_a} \left(\frac{3 s_i}{\eta_i} \mathbf{1}\{b_i = (P_a)_i\} \right). \tag{7}$$

It is an unbiased local-Pauli shadow feature, with the independent-noise correction written explicitly. This measurement primitive is not claimed new; see the classical-shadow framework [3] and noisy-shadow work [4].

Let $p_h(b,s) = \operatorname{Tr} \rho_h E_{b,s}$, $m_a = \mathbb{E}_h Y_a = \operatorname{Tr} \rho_h P_a$, and

$$V = \operatorname{Cov}_{p_h}(Y), \qquad F_M = \mathbb{E}_h\big[(\nabla_h \log p_h)(\nabla_h \log p_h)^T\big]. \tag{8}$$

Every $p_h(b,s)$ is positive. F_M is the classical Fisher matrix of the full recorded outcome, including the randomized bases.

The SLD quantum Fisher matrix F_Q is defined by Hermitian L_a satisfying

$$\partial_{h_a}\rho = \tfrac{1}{2}(\rho L_a + L_a \rho), \qquad (F_Q)_{ac} = \tfrac{1}{2}\operatorname{Tr}\rho\{L_a, L_c\}. \tag{9}$$

It concerns parameters encoded in the *equilibrium state family* (2), not a unitary encoding $e^{-itA}\rho e^{itA}$. Each directional $u^T F_Q u$ bounds every measurement's information. No simultaneous attainability of the multiparameter SLD Cramer–Rao matrix is assumed.

3 Main theorem: a size-independent readout-loss matrix

Set

$$d_{\mathrm{ov}} = 1 + k(\Delta - 1), \tag{10}$$

$$\nu_{\mathrm{meas}} = d_{\mathrm{ov}}\left[\left(\frac{3}{\eta^2}\right)^k - 1\right], \tag{11}$$

$$w_0 = 4gk(k+\ell)\Delta, \tag{12}$$

$$\nu = \nu_{\mathrm{meas}} + \frac{\beta}{12} w_0 = \nu_{\mathrm{meas}} + \frac{\beta g}{3} k(k+\ell)\Delta. \tag{13}$$

These constants are conservative and are not asserted optimal.

Theorem 3.1 (Local-readout information loss)**.** *Under* (1)–(3) *and* (6)*,*

$$0 \preceq V - C \preceq \nu I_m. \tag{14}$$

Moreover,

$$\beta^2 C(C + \nu I_m)^{-1} C \preceq \beta^2 C V^{-1} C \preceq F_M \preceq F_Q \preceq \beta^2 C. \tag{15}$$

In particular,

$$0 \preceq \beta^2 C - F_M \preceq \beta^2 (V - C) \preceq \beta^2 \nu I_m, \tag{16}$$

$$0 \preceq F_Q - F_M \preceq \beta^2 \nu I_m. \tag{17}$$

The conclusions are uniform in n, m at fixed $\beta, g, k, \ell, \Delta, \eta$. They do not require a lower bound on the smallest eigenvalue of C.

Full rank and the absence of linear dependence modulo the identity imply $C \succ 0$. Equation (14) therefore makes V^{-1} well defined. An extension to dependent dictionaries can instead be formulated on the quotient by identity and exact linear relations; it is not needed here.

3.1 Unbiased readout and the sparse covariance excess

Lemma 3.2 (Readout covariance defect)**.** *Let $N = V - V_q$. Then*

$$0 \preceq N \preceq \nu_{\mathrm{meas}} I_m. \tag{18}$$

For disjoint S_a, S_c, $N_{ac} = 0$ exactly, at every state ρ.

Proof. At one site,

$$\sum_{b,s} \frac{I + \eta_i s \sigma_b}{6} \frac{3s}{\eta_i} \mathbf{1}\{b = p\} = \sigma_p.$$

Taking tensor products proves $\sum_y E_y Y_a(y) = P_a$, hence unbiasedness.

For any real u let $f(y) = u \cdot Y(y)$, $A = A_u$. The positive-operator identity

$$\sum_y (f(y)I - A) E_y (f(y)I - A) = \sum_y f(y)^2 E_y - A^2 \succeq 0 \tag{19}$$

uses only $\sum E_y = I$ and $\sum f E_y = A$. Taking the state expectation and subtracting the common squared mean gives $u^T N u \geq 0$.

If the Pauli axes of P_a, P_c agree on their common support, direct tensor-product evaluation gives

$$\mathbb{E}(Y_a Y_c) = \left(\prod_{i \in S_a \cap S_c} \frac{3}{\eta_i^2} \right) \mathrm{Tr}(\rho P_a P_c). \tag{20}$$

If they disagree at any common site, $Y_a Y_c$ is identically zero. The symmetrized Pauli product expectation has magnitude at most one, and is zero when the strings anticommute. Thus $N_{ac} = 0$ for disjoint supports, while every entry has magnitude at most $(3/\eta^2)^k - 1$.

For fixed a, there are at most

$$1 + \sum_{i \in S_a} (\#\{c : i \in S_c\} - 1) \leq 1 + k(\Delta - 1)$$

indices with overlapping support, counting a. The row-sum bound on the real symmetric N proves its upper operator bound. Long-range correlations remain in both V and V_q and cancel in their difference. □

3.2 Thermal quantum variance and a local double-commutator bound

The next inequality is the established Brooks–Harris inequality in the normalization used here; see [1]. The variance-minus-susceptibility quantity also has the established interpretation of quantum variance [2]. We give its finite-dimensional proof to fix constants and assumptions.

Lemma 3.3 (Thermal covariance difference)**.** *For a Hermitian A, let $C(A, A)$ denote (4). Then*

$$0 \leq \mathrm{Var}_\rho(A) - C(A, A) \leq \frac{\beta}{12} \mathrm{Tr}\, \rho [A, [H, A]]. \tag{21}$$

The double-commutator expectation is nonnegative in a Gibbs state.

Proof. In an energy eigenbasis set $p_i = e^{-\beta E_i}/Z$,

$$L_{ij} = \begin{cases} (p_i - p_j)/(\log p_i - \log p_j), & p_i \neq p_j, \\ p_i, & p_i = p_j, \end{cases} \quad T_{ij} = \frac{p_i + p_j}{2}, \quad x_{ij} = \frac{\log p_i - \log p_j}{2}.$$

The BKM and symmetrized-variance weights are L_{ij} and T_{ij}, respectively, with centered A. Since

$$T_{ij} = L_{ij} x_{ij} \coth x_{ij}, \qquad 0 \le x \coth x - 1 \le x^2/3,$$

the difference is nonnegative and bounded by $L_{ij} x_{ij}^2/3$ termwise. The scalar inequalities hold at zero by continuity. For $x > 0$, the upper inequality is equivalent to

$$(3 + x^2) \sinh x - 3x \cosh x \ge 0.$$

The left side vanishes at zero and its derivative is $x(x \cosh x - \sinh x) \ge 0$; the latter parenthesis has derivative $x \sinh x \ge 0$. The lower inequality follows similarly from $x \cosh x \ge \sinh x$.

Direct expansion gives

$$\operatorname{Tr} \rho [A, [H, A]] = \sum_{ij} (p_i - p_j)(E_j - E_i) |A_{ij}|^2 = \sum_{ij} \frac{4 x_{ij}^2}{\beta} L_{ij} |A_{ij}|^2. \tag{22}$$

Centering does not change off-diagonal entries or commutators. Summing proves the assertion. □

Lemma 3.4 (Coefficient-space locality)**.** *Define the real symmetric matrix W by*

$$u^T W u = \operatorname{Tr} \rho [A_u, [H, A_u]].$$

Under (1) *and* (3)*,*

$$0 \preceq W \preceq w_0 I_m, \qquad w_0 = 4gk(k + \ell)\Delta. \tag{23}$$

Proof. Nonnegativity follows from (22). Expand the quadratic form in u and H_X. The term $[P_a, [H_X, P_c]]$ has norm at most $4 \|H_X\|$ and vanishes unless

$$X \cap S_c \ne \varnothing, \qquad S_a \cap (X \cup S_c) \ne \varnothing. \tag{24}$$

Let K_{ca} be the sum of $4 \|H_X\|$ over those X satisfying (24). Then $u^T W u \le |u|^T K |u|$.

For a fixed c, $\sum_{X : X \cap S_c \ne \varnothing} \|H_X\| \le kg$. Each such X has at most $(k + \ell)\Delta$ candidate supports meeting $X \cup S_c$. Therefore every row sum of K is at most w_0.

For a fixed a, split the last condition of (24) into $X \cap S_a \ne \varnothing$ or $S_c \cap S_a \ne \varnothing$, allowing double counting. In the first case, at most $\ell\Delta$ supports S_c meet each X, and the total weight of X touching S_a is at most kg. This contributes at most $4kg\ell\Delta$. In the second case there are at most $k\Delta$ choices of c, each with incident weight at most kg. This contributes at most $4k^2 g\Delta$. Column sums are consequently also bounded by w_0. The elementary row/column Schur bound gives $\|K\|_{2 \to 2} \le w_0$, proving the result. □

Lemmas 3.3–3.4 give

$$0 \preceq V_q - C \preceq \frac{\beta}{12} W \preceq \frac{\beta}{12} w_0 I_m. \tag{25}$$

Adding (18) proves (14). Neither covariance itself needs a bounded maximum eigenvalue.

3.3 Converting covariance control into actual readout information

Lemma 3.5 (Thermal response and moment information)**.** *For the measurement above,*

$$\partial_{h_b} m_a = -\beta C_{ab}, \qquad F_M \succeq \beta^2 C V^{-1} C, \qquad F_M \preceq F_Q \preceq \beta^2 C. \tag{26}$$

Proof. The differentiated exponential identity

$$\partial_{h_b} e^{-\beta H} = -\beta \int_0^1 e^{-\beta t H} P_b e^{-\beta(1-t)H}\, dt$$

and differentiation of the normalization give $\partial_{h_b} m_a = -\beta C_{ab}$.

Let $s = \nabla_h \log p_h$ be the outcome score. Since $\mathbb{E}s = 0$ and the features do not depend on h,

$$\mathbb{E}[(Y-m)s^T] = \nabla_h m = -\beta C.$$

The covariance matrix of $(Y-m, s)$ is positive semidefinite. Taking its Schur complement in V yields $F_M \succeq \beta^2 C V^{-1} C$. This is an information inequality for a specified physical measurement, not an assumed likelihood algorithm.

For completeness, in any real direction the SLD measurement inequality follows from Cauchy–Schwarz:

$$(\partial_u p_y)^2 = \left(\operatorname{Re}\operatorname{Tr} E_y \rho L_u\right)^2 \le p_y \operatorname{Tr} E_y L_u \rho L_u.$$

Sum after dividing by p_y to obtain $u^T F_M u \le \operatorname{Tr} \rho L_u^2 = u^T F_Q u$.

In the energy basis, the state derivative in direction u has entries $-\beta L_{ij}(A_u)_{cij}$, while the SLD entries divide these by T_{ij}. Thus the directional SLD quadratic form has weights $\beta^2 L_{ij}^2/T_{ij}$. Since the logarithmic mean is at most the arithmetic mean, these weights are at most $\beta^2 L_{ij}$. This proves the final inequality. □

Completion of Theorem 3.1. Write $R = V - C$, so $0 \preceq R \preceq \nu I_m$. Inversion reverses positive-definite order, yielding

$$CV^{-1}C \succeq C(C+\nu I_m)^{-1}C.$$

More sharply, without assuming C commutes with R,

$$C(C+R)^{-1}C - (C-R) = R(C+R)^{-1}R \succeq 0. \tag{27}$$

Consequently $\beta^2 C - F_M \preceq \beta^2 R$. The other sides follow from Lemma 3.5. This proves all claims. □

Remark 3.6 (Why the result does not contradict quantum enhancement). The upper loss is additive, not a small relative error in every direction. If $u^T C u = O(\|u\|^2)$, the theorem permits a substantial relative readout loss. Also F_Q here measures equilibrium encoding. It is distinct from the unitary-generator QFI commonly used as an entanglement witness. No assertion that multipartite quantum resources are generally useless in metrology follows.

4 A measurable certificate and an explicit local sensing procedure

4.1 Superextensive information becomes locally readable

Corollary 4.1 (Asymptotically complete local readout)**.** *For a sequence satisfying the same fixed local bounds and fixed β, η, suppose*

$$\frac{u_n^T F_Q u_n}{\|u_n\|_2^2} \longrightarrow \infty.$$

Then

$$\frac{u_n^T F_M u_n}{u_n^T F_Q u_n} \longrightarrow 1. \tag{28}$$

Proof. Divide (17) in direction u_n by $u_n^T F_Q u_n$. □

For a uniform collective direction with $\|u_n\|^2 = O(n)$, information growing faster than n therefore has the same leading scaling and coefficient under this fixed product readout as under the optimal quantum directional measurement. This statement is conditional on the enhanced scaling actually occurring. It does not assert that every phase or model has it. If temperature or local bounds change with size, the explicit $\beta^2\nu$ dependence must be retained rather than treating it as a constant.

4.2 A scalar moment, not the full likelihood

Fix one direction and consider $H(t) = H_0 + tA_u$, with all other parameters held fixed. Let

$$f = u \cdot Y, \quad V_u = \mathrm{Var}(f), \quad C_u = u^T C u, \quad q = \|u\|_2^2 .$$

Then

$$C_u \le V_u \le C_u + \nu q, \qquad \frac{d}{dt}\mathbb{E}f = -\beta C_u. \tag{29}$$

The information supplied by the first moment, in the usual local error-propagation sense, is

$$F_{\mathrm{mom}} = \frac{\beta^2 C_u^2}{V_u} \le u^T F_M u \le u^T F_Q u \le \beta^2 C_u. \tag{30}$$

Thus

$$\frac{F_{\mathrm{mom}}}{u^T F_Q u} \ge \frac{C_u}{C_u + \nu q}. \tag{31}$$

Only a scalar mean statistic is needed to reach asymptotic optimality in enhanced directions.

This has an explicit estimator interpretation when local calibration is supplied. At a reference t_0, write $\mu_0 = \mathbb{E}_{t_0} f$ and $\mu_0' = -\beta C_u(t_0)$. For N independent copies set

$$\widehat{t} = t_0 + \frac{\overline{f} - \mu_0}{\mu_0'}. \tag{32}$$

At t_0, this estimator is unbiased and its mean has derivative one. Its variance is exactly

$$\mathrm{Var}_{t_0}(\widehat{t}) = \frac{V_u}{N\beta^2 C_u^2}, \qquad 1 \le N(u^T F_Q u)\,\mathrm{Var}(\widehat{t}) \le 1 + \frac{\nu q}{C_u}. \tag{33}$$

The lower inequality is the scalar quantum Cramer–Rao bound; the upper follows from (29). This is *local* estimation. Calibration acquisition and a neighborhood in which a linear approximation is accurate must be paid for; no uniform finite-displacement reconstruction guarantee is implied.

4.3 A variance-only certificate without a forward Gibbs calculation

Corollary 4.2 (Variance certificate)**.** *Let V_u be the variance of the measured scalar f, and put $a = \nu q$. Then*

$$\beta^2 \frac{(V_u - a)_+^2}{V_u} \le F_{\mathrm{mom}} \le u^T F_M u \le u^T F_Q u \le \beta^2 V_u. \tag{34}$$

The left side is defined as zero when $V_u = 0$. Also

$$\beta\,(V_u - a)_+ \le -\frac{d}{dt}\mathbb{E}f \le \beta V_u. \tag{35}$$

Proof. Equation (29) implies $(V_u - a)_+ \le C_u \le V_u$. Insert these bounds in (30) and the response identity. □

The certificate needs a known perturbation direction, temperature, structural locality/strength bounds, and detector calibration. It does *not* require knowing the full background Hamiltonian, evaluating Z, solving for an SLD, perturbing the parameter experimentally, or integrating real-time response functions. If V_u/a is large, its lower and upper endpoints have ratio $(1 - a/V_u)^2$ close to one. This certifies near-optimal local readout and bounds the static susceptibility from ordinary records. It is not a state-agnostic QFI certificate: the Gibbs-family premise is essential.

Proposition 4.3 (Finite-copy confidence interval)**.** *Let $p \in (0,1)$, $\zeta > 0$, and*

$$L_u = (3/\eta)^k \, \|u\|_1 \, .$$

On $2M$ independent copies form

$$Z_j = \frac{(f_{2j-1} - f_{2j})^2}{2}, \qquad \widehat{V} = \frac{1}{M} \sum_{j=1}^{M} Z_j.$$

If

$$M \geq \frac{2L_u^4}{\zeta^2} \log \frac{2}{p}, \tag{36}$$

then $|\widehat{V} - V_u| \leq \zeta$ with probability at least $1 - p$. Put $V_- = \max\{0, \widehat{V} - \zeta\}$ and $V_+ = \widehat{V} + \zeta$. On that event,

$$\beta^2 \frac{(V_- - a)_+^2}{V_-} \leq u^T F_M u \leq u^T F_Q u \leq \beta^2 V_+, \tag{37}$$

with the same zero convention.

Proof. The feature bound gives $|f| \leq L_u$, so $0 \leq Z_j \leq 2L_u^2$. Pair independence gives $\mathbb{E}Z_j = \mathrm{Var}(f)$. Hoeffding's inequality yields

$$\Pr(|\widehat{V} - V_u| > \zeta) \leq 2 \exp\left(-\frac{M\zeta^2}{2L_u^4}\right).$$

For $v \geq 0$, $(v-a)_+^2/v$ (zero at $v = 0$) is nondecreasing: it is zero below a, and its derivative above a is $1 - a^2/v^2 \geq 0$. Apply Corollary 4.2. □

This shot bound is deliberately conservative, not an optimal statistical result. At fixed k, Δ, one recorded copy takes n one-qubit measurements and $O(n + km)$ arithmetic work to compute all features or the specified scalar. A streaming scalar certificate needs only running sums beyond storage of the supplied dictionary and coefficients. There are no entangling measurement gates, Gibbs-gradient queries, or partition-function computations. The preparation of $2M$ independent equilibrium copies is still a real experimental resource. Bias from preparation error, imperfect structural assumptions, correlated noise, unknown visibility, or temperature miscalibration is not covered by (36).

For example, if $L_u = O(n)$ and V_u is known to scale as $n^{1+\alpha}$, a fixed-relative-accuracy variance estimate has the conservative cost $O(n^{2-2\alpha} \log(1/p))$ pairs. This conditional scaling does not include equilibration time or establish that any particular device realizes such a regime.

5 What the previous covariance candidate would add

The exact original candidate assumes $H = \sum_a h_a P_a$, both (1) and incident weighted strength g, no background outside that bounded-incidence dictionary, and an unrestricted Gibbs state. It proposes

$$C \succeq cI_m, \qquad c = \exp[-C_{k,\Delta}(1 + \beta g)^2]. \tag{38}$$

Here $C_{k,\Delta}$ is a claimed finite constant, distinct from the covariance matrix C.

Corollary 5.1 (Explicitly conditional all-directions upgrade)**.** If (38) *holds, then in its exact setting (take $\ell = k$ above),*

$$F_M \succeq \frac{c}{c+\nu}\,\beta^2 C \succeq \frac{c}{c+\nu} F_Q, \qquad F_M \succeq \frac{\beta^2 c^2}{c+\nu} I_m. \tag{39}$$

Proof. For every eigenvalue $\lambda \geq c$, $\lambda^2/(\lambda+\nu) \geq [c/(c+\nu)]\lambda$. Apply functional calculus to (15), then use $F_Q \preceq \beta^2 C$. □

This would say that one fixed noisy separable measurement is uniformly competitive with the quantum information metric in *every* parameter direction, not just highly informative directions. It is not used elsewhere in this note and is not promoted to an established unconditional result.

A targeted reconstruction of the predecessor's key step found no decisive defect in the exact-adjoint-before-truncation ordering: for local maps $\mathcal{G}_i$ exactly self-adjoint in the KMS inner product,

$$\langle \mathcal{G}_i P_a, \mathcal{G}_j P_c \rangle_K = \langle P_a, \mathcal{G}_i \mathcal{G}_j P_c \rangle_K.$$

Only then are local approximations inserted. Disjoint approximate supports annihilate the product, so the estimate need not bound $\mathrm{Cov}_\rho(P_a, P_c)$. The former manuscript's coefficient counting and choice of spatial decay exponent can absorb bounded-degree volume growth. Nevertheless, the Gaussian locality, joint frequency/spatial tails, complete constants, and combined universal theorem remain subject to independent proof reconstruction. A targeted recheck is not external verification or Lean formalization.

Most importantly, the auxiliary map need not generate a completely positive semigroup. An information inequality for it does not produce a physical rapidly mixing Gibbs sampler.

6 Decisive limitations and counterexamples

6.1 The extensive order of loss is unavoidable for this fixed readout

Take independent qubits with $H(h) = \sum_i h_i Z_i$ at $h = 0$, with common visibility η. The SLD Fisher matrix for the h_i is

$$F_Q = \beta^2 I_n.$$

In the fixed random-Pauli measurement, only the Z basis (probability 1/3) detects the local derivative, and detector visibility multiplies the score by η. Hence

$$F_M = \frac{\beta^2 \eta^2}{3} I_n. \tag{40}$$

For the collective direction $u = (1, \ldots, 1)$, the loss is $\beta^2(1 - \eta^2/3)n$. A universal subextensive loss bound for this fixed protocol is false. This is not evidence that entangling measurements are necessary in the example: measuring every qubit in Z is already optimal when that direction is selected in advance.

6.2 Independent detector noise cannot be replaced by a marginal-visibility condition

The following fully explicit family satisfies the assumptions of the main theorem on the Hamiltonian and parameter dictionary, but not its detector-independence hypothesis. Consider

$$H_n(t) = -\frac{J}{n} \sum_{i<j} Z_i Z_j + t M_n, \qquad M_n = \sum_i Z_i. \tag{41}$$

The individual parameter dictionary may be the single-site Z_i, with $k = \Delta = 1$; t is the all-ones direction. The background has $\ell = 2$ and incident weighted strength at most J at $t = 0$.

At $t = 0$ the state is spin-flip symmetric and $\mathbb{E}M_n = 0$. A configuration whose minority spin count is $r \in [n/4, n/2]$ has energy at least $3Jn/8$ above an aligned configuration, since its energy penalty is $(2J/n)r(n-r)$. Counting at most 2^n configurations and retaining the two aligned configurations in the partition function gives

$$\Pr(|M_n| \le n/2) \le 2^{n-1} e^{-3\beta Jn/8}.$$

For fixed $3\beta J/8 > \log 2$, therefore,

$$\operatorname{Var}(M_n) \ge \frac{n^2}{4}\left(1 - 2^{n-1}e^{-3\beta Jn/8}\right) = \Omega(n^2). \tag{42}$$

Because the parameter commutes with the Hamiltonian, $F_Q(t) = \beta^2 \operatorname{Var}(M_n) = \Theta(n^2)$ at zero.

Now let one common detector bit, with probability $p \in (0, 1/2)$, reverse *all* recorded signs on a copy. Every individual site has visibility $\eta = 1 - 2p > 0$, but its noise is correlated with every other site. For this diagonal model the clean record law obeys $q_t(b, -s) = q_{-t}(b, s)$: non-$Z$ outcomes are conditionally fair, and the diagonal Gibbs distribution is reversed by a global spin flip. The noisy law is

$$\widetilde{q}_t = (1-p)q_t + pq_{-t}.$$

At zero, $\widetilde{q}_0 = q_0$ while $\partial_t \widetilde{q}_0 = \eta \partial_t q_0$. Consequently

$$F_{M,\mathrm{common}}(0) = \eta^2 F_{M,\mathrm{clean}}(0) \le \eta^2 F_Q(0). \tag{43}$$

The information loss is at least $(1-\eta^2)F_Q(0) = \Omega(n^2)$, not $O(n)$. Thus positive per-site marginal visibility alone is insufficient for the theorem. No claim about preparing this finite-temperature phase mixture rapidly is made.

6.3 Strong covariance conditioning does not make observed likelihood convex

Let a single qubit have

$$H(x, z) = xX + zZ, \quad r = \sqrt{x^2 + z^2}, \quad f(r) = \tanh(\beta r)/r.$$

Conditioned on measuring in the Z basis, the $+1$ probability is $(1 - zf(r))/2$. The negative log likelihood for that observation, up to a basis-selection constant, is

$$\mathcal{L}(x, z) = \log 2 - \log(1 - zf(r)).$$

At $x > 0, z = 0$,

$$\nabla^2 \mathcal{L} = \begin{pmatrix} 0 & f'(x) \\ f'(x) & f(x)^2 \end{pmatrix}, \qquad \det \nabla^2 \mathcal{L} = -f'(x)^2 < 0, \tag{44}$$

where

$$f'(x) = \frac{\beta x \operatorname{sech}^2(\beta x) - \tanh(\beta x)}{x^2} < 0.$$

The strict sign follows by differentiating $\tanh y - y \operatorname{sech}^2 y$, whose derivative is $2y \operatorname{sech}^2 y \tanh y > 0$ for $y > 0$.

This example lies in a bounded one-site parameter domain with positive thermal covariance. It decisively disproves the inference "strongly convex log partition function implies a convex observed-data likelihood for local quantum measurement." It does not prove computational hardness of all fitting algorithms or rule out a different convex formulation.

6.4 Other scope boundaries

The information comparisons are local in parameter space. They do not directly bound finite-displacement testing error, prove global optimization convergence, or solve nuisance-parameter identifiability. Unknown temperature can be confounded with an overall Hamiltonian scale. Exact fixed-charge restrictions can have null parameter directions. At zero temperature full rank is lost. Preparation errors and correlated sampling can invalidate the certificate. None of these interfaces is silently included.

7 Relationship to prior work and scientific significance

The proof deliberately separates established ingredients from the assembled theorem. The static-variance/BKM comparison is the Brooks–Harris inequality, reviewed and generalized in [1]; quantum variance is an established object [2]. Random Pauli shadow measurements and their use for local observables are established [3]; noisy variants are also established [4]. Thermal Hamiltonian-parameter QFI bounds were studied directly in [5]. The moment-score Schur complement and SLD measurement bound are standard information inequalities, with proofs above.

The specific result established here is an *additive, coefficient-matrix, size-independent information-loss bound for a specified parameter-independent noisy product readout*, together with a variance-only operational certificate. The bounded covariance *differences* are central. This differs from merely lower bounding optimal QFI, estimating unitary-encoding QFI, measuring only a small subsystem, or proving Hamiltonian identifiability.

The targeted search did not locate this exact complete statement. That is not an exhaustive novelty or priority clearance. In particular, an inequality obtained by combining established results can be independently useful without being a new foundational principle in the historical sense.

There are already arbitrary-temperature Hamiltonian-learning algorithms [6], and near-optimal local lattice-learning results [7]. September 2026 quantum score matching provides a distinct learning construction [8]; October 2026 stationarity tests address Gibbs and metastable-state learning [9]. Their algorithmic assumptions, parameter-dependent or derived observables, and associated information functionals are not replaced by a classical-Fisher comparison. We claim neither the first efficient local-learning algorithm nor a computational improvement over them.

What is gained is a concrete answer to a different question: when equilibrium response is collectively enhanced, a blind local detector need not forfeit that enhancement, and the enhancement's accessibility can be certified from its records without a forward many-body calculation. Under a verified global covariance floor, the same theorem would further control readout efficiency for every direction. Without that floor, poorly conditioned directions remain poorly conditioned.

8 Reproducibility, provenance, and remaining obligations

The accompanying `finite_checks.py` performs 38 finite floating-point checks (seed 20261012): 36 two- to four-qubit Gibbs instances with noncommuting directions, additional background terms absent from the parameter dictionary, three inverse temperatures, and two detector visibilities; and two maximally mixed sharpness cases. It directly enumerates the product POVM, differentiates the Gibbs state, computes the measured and quantum information/covariance matrices, and tests the stated matrix inequalities. The declared tolerances passed. The smallest recorded inequality eigenvalue was approximately -2.79×10^{-16}, and the largest response-identity discrepancy was 8.44×10^{-15}. The exact nonconvex formula was also evaluated at $x = \beta = 1$.

The numerical tests do not prove the universal theorem, establish originality, or independently validate physical assumptions.

The input repository was pinned to Astra revision

`142e27d7eb0a0590a8fc649c63a95a818dca95aa`.

The task entry point, recent intake, and dossier `frontier/dossiers/N711.txt` were read. The original global manuscript was recovered from the mounted research ZIP; its LaTeX SHA256 is

`40a4144b055c1bf96d17567caf921fbe237b2900e041f5fb2569c7ccccc12980`.

The repository also preserves it at

`updates/EB/package/global_thermal_information/recovered/global_thermal_information.tex`.

A targeted `openai/math` search returned routes at revision

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

No theorem from that repository was used as a proof premise. Search results or archival status are not mathematical verification. No repository write or publication was performed.

The main information-loss theorem and certificate above have complete finite-dimensional derivations independent of N711. They still require external proof review and priority assessment. Uniform conditioning for the smallest-information directions remains conditional on the earlier candidate; generic efficient equilibrium preparation and globally efficient parameter fitting remain separate obligations. The certificate procedure itself has explicit measurement and arithmetic costs, but no physical speedup is claimed without paying for independent Gibbs copies.

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